

7(a)(i)	light not absorbed / stop photosynthesis / stop photoactivation / stop light dependent stage ;	1
---------	---	----------

Question	Answer	Marks								
7(a)(ii)	any three from: 1 all decrease in absorbance (with time) ; 2 (all decrease) at constant rate ; 3 green, highest absorbance (throughout) / absorbance stays fairly constant / smallest decrease in absorbance ; ora red 4 smallest rate of decrease in green ; ora red 5 data quote – 2 colours compared to support mp1 or mp3 ; <table><tr><td></td><td>4 mins</td></tr><tr><td>green</td><td>1.44</td></tr><tr><td>blue</td><td>0.90</td></tr><tr><td>red</td><td>0.70</td></tr></table>		4 mins	green	1.44	blue	0.90	red	0.70	3
	4 mins									
green	1.44									
blue	0.90									
red	0.70									
7(a)(iii)	any three from: with red light ora green light 1 more, light / energy, absorbed (by chlorophyll) ; 2 (so) more, photoactivation / electrons emitted ; 3 more / faster rate of, electrons transferred to DCPIP ; 4 more / faster, decolourisation of DCPIP ;	3								

Question	Answer					Marks																								
7(b)	<table><tr><td>factor</td><td>stage</td><td>✓ or ✗</td><td rowspan="2">}</td><td rowspan="2">;</td></tr><tr><td rowspan="2">carbon dioxide concentration</td><td>Calvin cycle</td><td>✓</td></tr><tr><td>photophosphorylation</td><td>✗</td></tr><tr><td rowspan="2">light intensity</td><td>Calvin cycle</td><td>✗</td><td rowspan="2">}</td><td rowspan="2">;</td></tr><tr><td>photophosphorylation</td><td>✓</td></tr><tr><td rowspan="2">temperature</td><td>Calvin cycle</td><td>✓</td><td rowspan="2">}</td><td rowspan="2">;</td></tr><tr><td>photophosphorylation</td><td>✓</td></tr></table>					factor	stage	✓ or ✗	}	;	carbon dioxide concentration	Calvin cycle	✓	photophosphorylation	✗	light intensity	Calvin cycle	✗	}	;	photophosphorylation	✓	temperature	Calvin cycle	✓	}	;	photophosphorylation	✓	3
factor	stage	✓ or ✗	}	;																										
carbon dioxide concentration	Calvin cycle	✓																												
	photophosphorylation	✗																												
light intensity	Calvin cycle	✗	}	;																										
	photophosphorylation	✓																												
temperature	Calvin cycle	✓	}	;																										
	photophosphorylation	✓																												

Question	Answer				Marks
----------	--------	--	--	--	-------